

Exercițiul 4

- a) Fie $\{f_n\} \subset F(A, \mathbb{R})$ pentru care $f_n: A \rightarrow \mathbb{R}$. f_n converge punctual către o funcție f (not: $f_n \rightarrow f$) pe A dacă toate graficele funcției ~~se intersectează~~ pot fi restrinse la un singur punct.

$$\Leftrightarrow \lim_{n \rightarrow \infty} \alpha_n = 0, \text{ unde } \alpha_n = \|f_n - f\|_{\infty} \text{ reprezintă norma}$$

$$\Leftrightarrow \alpha_n = \sup_{x \in \mathcal{C}} |f_n(x) - f(x)|$$

Exemplu: $f_n: (0, \infty) \rightarrow \mathbb{R}, f_n(x) = \frac{5}{n^2} \sqrt{x}$

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{5}{n^2} \sqrt{x} = 0 \quad \forall x \in (0, \infty)$$

$$\Rightarrow f_n \xrightarrow{(0, \infty)} f, \quad f(x) = 0$$

b) $f_n \in F(\mathbb{R}, \mathbb{R}), \quad f_n: \mathbb{R} \rightarrow \mathbb{R}$

$$f_n(x) = \frac{3n^2 x}{(1+n^2 x^2)^2} \quad \forall x \in \mathbb{R}$$

Fie $x \in \mathbb{R}$ arbitrar ales:

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{3n^2 x}{(1+n^2 x^2)^2} = \lim_{n \rightarrow \infty} \frac{3n^2 x}{1+2n^2 x^2+n^4 x^4} = 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow f_n \rightarrow f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = 0, \quad \mathcal{C} = \mathbb{R}$$

$$\begin{aligned} \text{Fie } \alpha_n &= \|f_n - f\|_{\infty} = \sup_{x \in \mathcal{C}} |f_n(x) - f(x)| = \sup_{x \in \mathbb{R}} \left| \frac{3n^2 x}{(1+n^2 x^2)^2} \right| = \\ &= \sup_{x \in \mathbb{R}} \frac{3n^2 |x|}{(1+n^2 x^2)^2} \end{aligned}$$

Fie $g: \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = \frac{3m^2x}{(1+m^2x^2)^2}$ o funcție auxiliară
careia îi studiem variația

$$g'(x) = \left(\frac{3m^2x}{(1+m^2x^2)^2} \right)' = \frac{-}{-}$$

$$= \frac{(3m^2x)' \cdot (1+m^2x^2)^2 - ((1+m^2x^2)^2)' \cdot 3m^2x}{(1+m^2x^2)^4} =$$

$$= \frac{(1+m^2x^2)^2 - 2(1+m^2x^2) \cdot (1+m^2x^2)' \cdot 3m^2x}{(1+m^2x^2)^4} =$$

$$= \frac{(1+m^2x^2)^2 - 2(1+m^2x^2) \cdot 2x \cdot 3m^2x}{(1+m^2x^2)^4} =$$

$$= \frac{\cancel{(1+m^2x^2)} (1+m^2x^2 - 2 \cdot 2x \cdot 3m^2x)}{(1+m^2x^2)^4 \cdot 3} =$$

$$= \frac{1+m^2x^2 - 12m^2x^2}{(1+m^2x^2)^3} =$$

$$= \frac{1-11m^2x^2}{(1+m^2x^2)^3}$$

$$g'(x) = 0 \Leftrightarrow 1-11m^2x^2 = 0 \Leftrightarrow 1 = 11m^2x^2 \Leftrightarrow$$

$$\Leftrightarrow x^2 = \frac{1}{11m^2}$$

$$\Leftrightarrow x = \pm \frac{1}{m\sqrt{11}}$$

x	$-\infty$	$-\frac{1}{m\sqrt{11}}$	0	$\frac{1}{m\sqrt{11}}$	∞			
$g'(x)$	-	-	0	+	+	0	-	-
$g(x)$	$\searrow$		$\nearrow$		$\searrow$			
$ g(x) $	$\rightarrow$		$ g(-\frac{1}{m\sqrt{11}}) $	$\searrow$	$ g(0) $	$\nearrow$	$ g(\frac{1}{m\sqrt{11}}) $	$\searrow$

$$\begin{aligned}
 |g(-\frac{1}{n\sqrt{n}})| &= \left| \frac{3n^2 \cdot (-\frac{1}{n\sqrt{n}})}{(1+n^2 \cdot (-\frac{1}{n\sqrt{n}})^2)^2} \right| = \\
 &= \left| \frac{\frac{-3n}{\sqrt{n}}}{(1+n^2 \cdot \frac{1}{n^2})^2} \right| = \left| \frac{\frac{-3n}{\sqrt{n}}}{(1+\frac{1}{n})^2} \right| = \\
 &= \left| \frac{-3n}{\sqrt{n} \cdot (\frac{12}{11})^2} \right| = \left| \frac{-3}{\sqrt{n} \cdot (\frac{12}{11})^2} \right| \cdot |n|
 \end{aligned}$$

$$|g(0)| = 0$$

$$\Rightarrow \sup_{x \in \mathbb{R}} \frac{3n^2(x)}{(1+n^2x^2)^2} = \left| \frac{-3}{\sqrt{n} \cdot (\frac{12}{11})^2} \right| \cdot n$$

$$\Rightarrow \lim_{n \rightarrow \infty} Q_n = \frac{-3}{\sqrt{n} \cdot (\frac{12}{11})^2} \lim_{n \rightarrow \infty} n = \infty \neq 0 \Rightarrow$$

$$\Rightarrow f_n \not\rightarrow f$$